Hierarchical Reinforcement Learning and Real-World Adaptation for Loco-Manipulation

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Background

Legged loco-manipulators can approach, grasp, transport, and place objects, but completing these tasks reliably requires coordination between locomotion and arm motion. Hierarchical control can reuse established low-level locomotion or whole-body controllers while learning task-level actions. Recent studies demonstrate sampling-based model predictive control (SMPC) for demonstration generation, high-level reinforcement learning (RL), and simulation-to-real deployment [1, 2, 3]. The thesis aims to establish a reproducible sim-to-real learning framework for loco-manipulation and systematically evaluate how physical execution feedback can improve the reliability and data efficiency of high-level policy adaptation in multi-stage manipulation tasks.

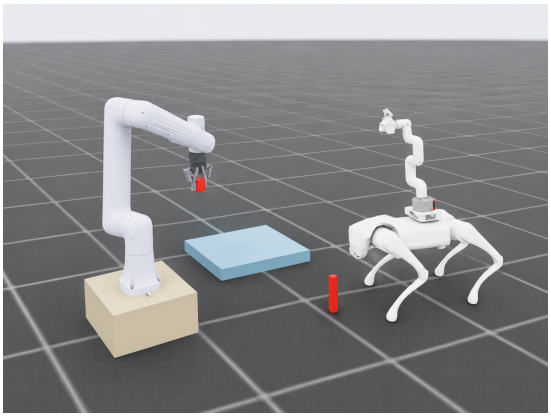


Figure 1: Example simulation at heterogeneous robotic manipulation.

An important challenge in deploying hierarchical loco-manipulation policies is maintaining reliable task execution under discrepancies between simulated and physical dynamics, perception, and contact interactions [3, 2]. In multi-stage manipulation tasks, uncertainties in grasp configuration, object pose estimation, contact friction, and actuator response can propagate across successive actions, reducing task completion reliability despite stable locomotion [3, 4]. Building upon hierarchical control architectures that combine sampling-based motion planning, reinforcement learning, and low-level motion control [1, 2], this thesis aims to establish a reproducible framework integrating simulation-based policy training, real-time robot control, and physical experimentation. Particular emphasis will be placed on characterizing sim-to-real performance degradation and systematically evaluating whether real-world interaction data can improve task success and policy adaptation efficiency [4].

The experimental platform consists of a **Unitree Go2 quadruped equipped with an Elephant Robotics myCobot 280 Pi 6-DoF manipulator**, serving as the primary learning and execution system. A separate **Dobot Nova 5** will support automated physical experimentation through programmable object presentation, repeatable interaction trajectories, and task resetting. Synchronized robot-state measurements and calibrated object-pose observations, where available, will support quantitative evaluation of manipulation performance and the collection of real-world experience for subsequent policy adaptation.

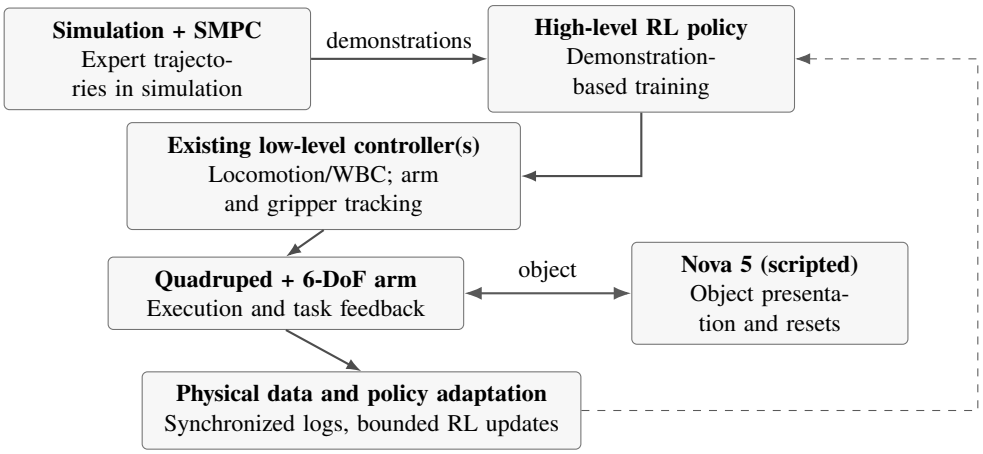


Figure 2: Proposed hierarchical training and physical evaluation framework. High-level learning is carried out for the mobile manipulator; the Nova 5 provides a programmable physical task environment.

Potential Research Questions (Select One)

- How can existing low-level locomotion or whole-body controllers (e.g., MPC or RL-based WBC) be integrated with hierarchical reinforcement learning and sampling-based motion planning (e.g., SMPC) to enable coordinated, long-horizon loco-manipulation on a quadruped equipped with a 6-DoF arm?
- How can high-level task objectives be translated into feasible base and end-effector trajectories while satisfying kinematic, dynamic, and environmental constraints?
- Which discrepancies between simulation and reality most strongly affect task performance, such as grasp-pose errors, contact changes, control delays, and object motion during transport?
- To what extent can real-world interaction data improve simulation-trained policies, and how do different adaptation strategies compare with zero-shot deployment in terms of task success, robustness, and data efficiency?

Deliverables

- Reproducible simulation scene, SMPC demonstration pipeline, high-level RL training configuration, and reusable low-level control interfaces.
- Real-robot policy deployment and data collection with a calibrated, repeatable manipulation task and documented safety constraints.
- Quantitative or qualitative evaluation of zero-shot transfer, failure mechanisms, and at least one bounded real-world adaptation approach against appropriate baselines.
- Source code, experimental protocol, thesis report, and demonstration video.

Scope and Team Structure

The MSc thesis prioritizes a functional research framework and careful evaluation, not the simultaneous development of new WBC, SMPC, hierarchical RL, and perception algorithms.

- **One student (Learning and real-robot deployment):** Justify and maintain available low-level controllers, implement/adapt a simulation control-to-RL pipeline, deploy one complete manipulation task, and compare zero-shot execution with one limited-data adaptation strategy. Nova 5 automation can initially be scripted with supervised resets.
- **Two students (Learning + physical experimental systems):** One focuses on simulation, demonstrations, high-level RL, and policy adaptation; the other focuses on physical robot integration, sensing, data collection, and sim-to-real evaluation. The extended scope may include more task variations and systematic generalization tests.

Student Background and Scope

- Strong interest in robotics, reinforcement learning, whole body control, and autonomous robotic systems.
- Proficiency in Python and C/C++, preferably with experience with reinforcement learning algorithms such as PPO, SAC, or TD3.
- Basic understanding of robot kinematics, dynamics, feedback control, and motion planning.
- Some experience with simulation environments such as Isaac Sim, Isaac Lab, Matlab Simulink.
- Willingness to run real robot experiments in cluttered environments, with the hardware effort this entails.
- Optional: Experience with ROS/ROS 2, embedded systems (sensor/actuator/controller), MPC, whole-body control, imitation learning, computer vision, or sim-to-real transfer.

Supervision and Resources

Proposed supervisors:

- **Xuezhi Niu**, PhD student, Department of Information Technology, Uppsala University
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Resources: CPS Lab facilities at Uppsala University; the available quadrupedal robot Go2 from Unitree and 6-DoF manipulators myCobot from Elephant Robotics, Dobot Nova 5, computing resources for simulation and policy training, ROS 2 development machines, and prototyping equipment. Access and task design are subject to hardware readiness, validated payload limits, and laboratory safety procedures.

Selected References

[1] RAI Institute, "Combining Sampling and Learning for Dynamic Whole-Body Manipulation," October 2025. https://rai-inst.com/resources/blog/combining-sampling-and-learning-for-dynamic-whole-body-manipulation/.

[2] M. Schuck *et al.*, "Learning Loco-Manipulation From SMPC Demonstrations With Sparse Offline-to-Online RL," *CoRL*, 2026. https://arxiv.org/abs/2608.12063.

[3] H. Zhang *et al.*, "SLIM: Sim-to-Real Legged Instructive Manipulation via Long-Horizon Visuomotor Learning," arXiv:2501.09905, 2025. https://arxiv.org/abs/2501.09905.

[4] R. Mendonca *et al.*, "Continuously Improving Mobile Manipulation with Autonomous Real-World RL," *CoRL*, 2024. https://arxiv.org/abs/2409.20568.
[5] M. Seo *et al.*, "LEGATO: Cross-Embodiment Imitation Using a Grasping Tool," *IEEE Robotics and Automation Letters*, vol. 10, no. 3, pp. 2854–2861, 2025. doi:10.1109/LRA.2025.3535182.
[6] A. Tulbure, C. Scheidemann, E. Steiner, and M. Hutter, "Task-Oriented Robot-Human Handovers on Legged Manipulators," *Proc. ACM/IEEE HRI*, 2026. doi:10.1145/3757279.3785629.
[7] Jin, Y. *et al.*, "Reinforcement Learning Based Multi-Terrain Adaptive Whole-Body Control for a Quadruped Manipulator," arXiv:2610.23548, 2026. https://arxiv.org/abs/2610.23548.

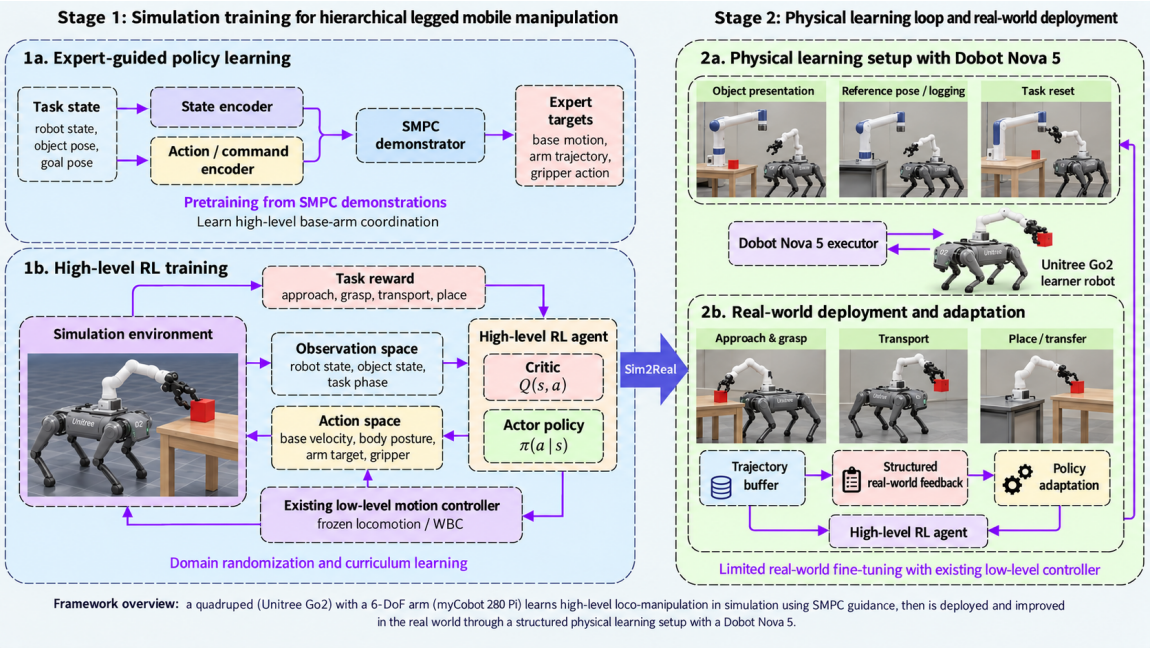


Figure 3: Example Hierarchical Reinforcement Learning and Real-World Adaptation Framework.